

AEXL: Enhancing Path Prediction with Active Explainable Learning via Large Language Models

Achref Doula
Telecooperation Lab (TK)
Technical University of Darmstadt
Darmstadt, Germany
doula@tk.tu-darmstadt.de

Max Mühlhäuser
Telecooperation Lab
Technical University of Darmstadt
Darmstadt, Germany
max@informatik.tu-darmstadt.de

Alejandro Sanchez Guinea
Telecooperation Lab (TK)
Technical University of Darmstadt
Darmstadt, Germany
sanchez@tk.tu-darmstadt.de

Abstract

Active learning (AL) reduces annotation costs by selectively querying humans for uncertain data samples, thus improving data efficiency. In traditional AL, annotators simply label the data without gaining a deeper understanding of the reasoning of the model. However, incorporating model-generated explanations can enhance annotation quality by helping annotators interpret decisions and detect errors, a crucial need in safety-critical tasks, such as path prediction, where reliable annotations are vital for autonomous navigation. We propose Active Explainable Learning (AEXL), a novel AL framework that integrates natural language explanations via large language models (LLMs), enabling annotators to interpret predicted paths and identify errors more effectively. Our preliminary evaluations indicate that AEXL reaches 99% of the anticipated performance using only 60% of the data. The results of our preliminary user study show that natural language explanations boost annotator understanding, trust, and reduce cognitive load compared to approaches that lack explanatory guidance. Our work aims to foster future research on explanation-driven AL to improve model performance and human-AI collaboration.

CCS Concepts

• **Computing methodologies** → **Machine learning**.

Keywords

Human-in-the-Loop Machine Learning, Explainable Machine Learning, Large Language Models, Autonomous Driving

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1 Introduction

Active learning (AL) is a machine learning paradigm that strategically selects the most informative data samples for annotation,

offering a cost-effective human-in-the-loop approach to reducing annotation overhead [16]. Compared to traditional supervised learning, AL stands out as an efficient strategy to reduce the substantial data requirements of training robust models. AL selectively identifies challenging samples from an unlabeled dataset based on specific criteria, directing these to human annotators for correction before incorporating them into the training set. By focusing on difficult or uncertain data, AL enhances model performance while requiring smaller datasets. Its effectiveness in improving training efficiency has been demonstrated in domains such as image classification [18, 27], natural language processing [9], and medical image analysis [3]. Despite its benefits, traditional AL offers little insight into the model’s reasoning process. This restriction is particularly problematic for safety-critical tasks, such as multimodal path prediction, where annotators must understand the model decision-making process to ensure reliable annotations and iteratively improve predictions.

Multimodal path prediction involves forecasting future trajectories of agents (e.g. vehicles, pedestrians) in urban environments based on observed states [1]. These predictions are vital for autonomous navigation, as they inform the planning and decision-making processes of vehicles and robots. However, the inherent uncertainty in agent behavior and the complexity of dynamic urban environments pose significant challenges. Models must capture diverse interactions and complex patterns [2, 13], which requires extensive, high-quality datasets for effective training. Collecting such datasets typically involves exhaustive driving campaigns across multiple cities, encompassing varied urban layouts, weather conditions, and agent behaviors, resulting in substantial financial and logistical costs [4, 6].

Although AL mitigates some of these costs by focusing on informative samples, it fails to address the interpretability needs of annotators. Explanatory Interactive Learning (XIL) [20] extends AL by incorporating model explanations, enabling annotators to understand and evaluate the reasoning behind predictions. This deeper interaction improves annotation quality and aids in identifying model errors. However, applying XIL to multimodal path prediction is challenging due to the opacity of path prediction models, which often function as black-box systems. Overcoming these challenges requires innovative approaches that balance predictive accuracy, interpretability, and human involvement.

In this work, we propose Active Explainable Learning (AEXL), a novel active learning framework that addresses these challenges in path prediction tasks. AEXL integrates neuro-symbolic reasoning with large language models (LLMs) to generate interpretable natural language explanations for path predictions. Using interpretable

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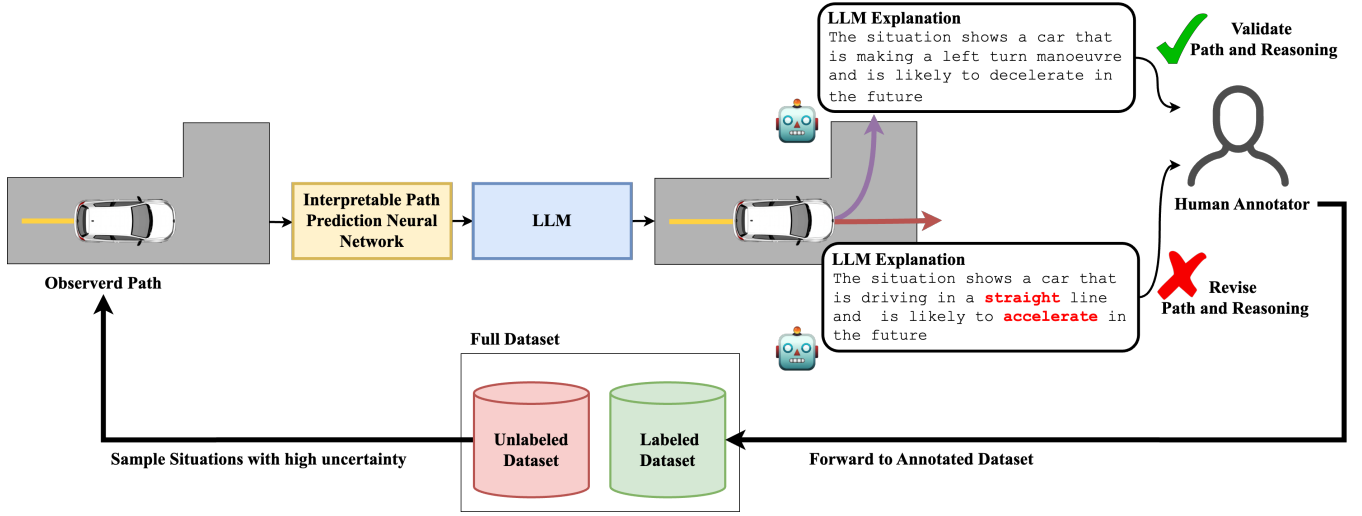


Figure 1: The AEXL framework for Active Explainable Learning. AEXL enhances active learning for path prediction by integrating neuro-symbolic reasoning with large language models (LLMs). A high uncertainty scenario is sampled from the unlabeled dataset. Observed path data is processed by a neuro-symbolic model to generate interpretable first-order logic rules describing predicted paths. These rules are then translated into natural language explanations by an LLM, offering human-readable justifications for predicted trajectories. Annotators are presented with both the predicted paths and their explanations, enabling them to check the paths and the reasoning, correct or validate them, and forward the corrected scenarios to the labeled dataset.

models such as [10], AEXL generates symbolic first-order logic rules that describe the predicted paths. These rules are then translated into human-readable natural language explanations by an LLM, as shown in Figure 1. This process empowers annotators to understand and correct predictions, thereby improving annotation quality and reducing cognitive effort. Additionally, AEXL incorporates a novel uncertainty heuristic to identify semantically inconsistent scenarios, prioritizing them for annotation and refining the model’s reasoning during iterative learning cycles.

We perform a preliminary evaluation of AEXL using the Argoverse benchmark [6], a widely recognized dataset in the autonomous driving research community. The results demonstrate that AEXL achieves 99% of the performance attainable with the full training dataset while using only 60% of the data. Furthermore, our preliminary user study reveals that natural language explanations significantly enhance interpretability, reduce cognitive load, and increase trust compared to baseline approaches lacking explanations. The preliminary findings show the importance of combining active learning with transparent, human-centered explanations to improve both model performance and human-AI collaboration.

2 Related Work

2.1 Active Learning for Data-Efficient ML

Active learning (AL) aims to reduce annotation overhead by selectively querying labels for the most informative data instances [11, 17]. Traditional AL strategies rely on uncertainty sampling, where the model requests labels for samples with high predictive entropy or variance. More recent approaches incorporate diversity or representativeness into the selection criterion, ensuring that

the labeled set spans a wide range of patterns within the dataset [14, 25]. Despite the demonstrated effectiveness of these methods in minimizing labeling costs, most AL pipelines focus on maximizing model accuracy while offering limited support for interpretability or user interaction.

2.2 Neuro-Symbolic Models for Interpretability

Neuro-symbolic architectures represent suitable candidates to be included in AL-based frameworks. They combine neural networks with symbolic representations to perform logic-based human interpretable reasoning [15, 22]. Neuro-symbolic models have been applied in several tasks, such as semantic image interpretation [8] and cognitive robotics [7]. In autonomous driving, neuro-symbolic models have been applied to context understanding [21], multi-modal path prediction [10], and in generating optimal navigation programs [19]. However, one recurring challenge is that symbolic representations can become complex or unwieldy for non-technical stakeholders. Furthermore, ensuring the correctness of learned rules often requires careful curation or manual verification, especially in safety-critical settings like autonomous driving.

2.3 Large Language Models and Natural Language Explanations

The rapid development of large language models (LLMs) has enabled machine-generated text that can offer human-readable justifications for model decisions [12, 28]. Recent works have explored using LLMs to translate technical outputs, such as attention weights or symbolic logic, into explanatory narratives [5, 24]. While LLM-based explanations can significantly improve user understanding,

the veracity and clarity of generated text remain open concerns. This is particularly relevant when explanations influence high-stakes decisions, necessitating careful vetting and feedback from domain experts.

3 Approach

3.1 Approach Overview

In this paper, we propose AEXL, an active learning framework for path prediction. Our framework enhances the active learning process by learning logical explanations of the path prediction process and prioritizing semantically inconsistent scenarios to be forwarded to the human annotator. AEXL extends the role of the human in active learning from simply providing the annotations to interacting with the model through interpretable logic-based rules, allowing for more transparency and trust. As depicted in Figure 2, AEXL consists of an off-the-shelf neuro-symbolic path prediction neural network and a small subset of the training set that is initially labeled. The labeled subset is used to train the neuro-symbolic path generation model $\mathcal{M}$, for which the goal is to learn a first-order logic rule that describes the maneuver performed by the dynamic agent during a particular scenario. After the training is finished, we use $\mathcal{M}$ to perform inference on the remaining part of the dataset. At this stage, $\mathcal{M}$ generates a set of K possible future paths $\hat{\mathcal{Y}}$ and a set of possible first-order logic rules, p for each of the generated paths. The generated rules formalize the link between the observed maneuver, X_i^{past} , and the predicted future maneuver, X_i^{future} , using a soft implication relationship. Once the paths and the rules are generated, we estimate the uncertainty of the base model to generate consistent paths with the neuro-symbolic rule generator using a scoring function S_{AL} that we design to capture the semantic inconsistencies between the generated paths and the generated rules. The scenarios are then ranked according to the scoring and the top- n_{query} paths along with their generated rules are forwarded to a human annotator. The human annotator visualizes the scene, the observed path, the generated paths, and their corresponding rules, and corrects the predictions and the explanations. As is the case for AL procedures, this process is repeated until the annotation budget B is exhausted or a desired performance is achieved.

3.2 Neuro-Symbolic Reasoning Module

To offer interpretability beyond raw outputs, we use off-the-shelf neuro-symbolic prediction models, such as [10] that derive first-order logic rules. Each rule encodes conditions relating the agent's past maneuvers to its predicted future actions, providing a structured format for explaining the model's inferences. For an observed trajectory $Past(X_i)$ and a predicted trajectory $Future(X_i)$, the module synthesizes rules of the form:

$$Past(X_i) \wedge PastManeuver(X_i) \rightarrow Future(X_i) \wedge FutureManeuver(X_i), \quad (1)$$

where $PastManeuver(X_i)$ and $FutureManeuver(X_i)$ denote the agent's maneuvers in the past and future, respectively.

3.3 Natural Language Explanations

While symbolic rules are more interpretable than neural network weights, they can still be inaccessible to non-technical annotators. We therefore employ an LLM to convert these rules into concise, human-readable statements. For example, a rule of the form

$$Past(X_i) \wedge LeftTurn(X_i) \rightarrow Future(X_i) \wedge Decelerate(X_i) \quad (2)$$

might be translated as: "If the agent is turning left, it is expected to decelerate in the future."

These explanations accompany the predicted paths and relevant scene information, assisting annotators in evaluating how well the predictions conform to the logical expectations.

3.4 Active Learning Strategy

AEXL uses an uncertainty-driven active learning loop to pinpoint the most informative scenarios. Specifically, we compute an overall score S_{AL} for each unlabeled instance, combining multiple factors:

$$S_{AL} = S_{SC} + E_m + E_{MM} + E_{PS}, \quad (3)$$

where

- S_{SC} : Semantic consistency-based score, reflecting how well predicted maneuvers align with the neuro-symbolic rules:

$$S_{SC} = - \sum_{m=1}^{|P_{maneuver}|} c_m \cdot \log(p_m), \quad (4)$$

with c_m as the ground-truth maneuver class and p_m the predicted probability.

- E_m : Entropy of the neuro-symbolic module's maneuver predictions:

$$E_m = - \sum_{m=1}^{|P_{maneuver}|} p_m \cdot \log(p_m). \quad (5)$$

- E_{MM} : Average squared deviation of the K predicted trajectories from the ground truth, quantifying prediction dispersion and scene complexity:

$$E_{MM} = \frac{1}{K} \sum_{k=1}^K \|\hat{X}_{k,final} - X_{final}\|^2, \quad (6)$$

where $\hat{X}_{k,final}$ is the k -th predicted path and X_{final} is the ground truth final path.

- E_{PS} : Uncertainty in the model's path probabilities:

$$E_{PS} = - \sum_{k=1}^K s_k \cdot \log(s_k), \quad (7)$$

where s_k is the probability assigned to path k .

High S_{AL} scores suggest a scenario that is both uncertain and semantically inconsistent, thus worthy of human attention. After training initial models on a labeled subset, we evaluate each unlabeled instance using S_{AL} , select the top n_{query} highest-scoring samples, and obtain annotations (including corrections and rationales). The newly annotated data are then merged into the labeled pool for subsequent training, progressively refining both the path prediction model and the neuro-symbolic rules. The active learning loop iteratively selects the top n_{query} scenarios with the highest S_{AL} .

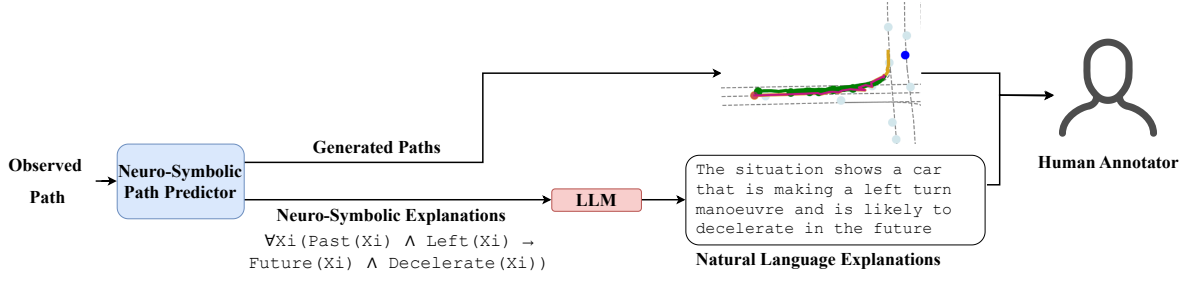


Figure 2: Explanatory process of AEXL. A neuro-symbolic path predictor generates predicted paths and logical explanations, which are translated into natural language by a large language model (LLM). A human annotator reviews these explanations to validate or refine the predictions, improving model interpretability and accuracy.

scores for annotation. Annotators review the predicted paths, natural language explanations, and scene context to provide corrections, which are incorporated into the training set.

4 Evaluation

In this section, we present the evaluation of our framework through two complementary studies: (1) the quantitative evaluation of the active learning approach and its impact on model performance, and (2) a user study to assess the interpretability, cognitive load, and trust engendered by the natural language explanations.

4.1 Active Learning Evaluation

4.1.1 Experimental Setup. To assess the efficiency of the active learning strategy, we used the Argoverse dataset [23], a benchmark for motion forecasting in autonomous driving. The dataset consists of over 30,000 scenarios recorded in urban environments, each capturing the positions of agents over time along with HD map information. The task is to predict future agent positions over a 3-second horizon based on a 2-second observation period.

We evaluated our approach using NeSyMoF [10], a state-of-the-art neuro-symbolic path prediction model, as the base model. We use Llama3.1 – 8B as the LLM to generate natural language explanations. The evaluation compared three settings including 2 baselines:

- (1) **Random Sampling (Baseline 1):** Scenarios are randomly selected for annotation without prioritization.
- (2) **Active Learning Without Semantic Consistency (Baseline 2):** Scenarios are prioritized using uncertainty measures but without the semantic consistency term in the scoring function.
- (3) **Active Learning With Semantic Consistency (Ours):** Scenarios are prioritized using the full scoring function, including semantic consistency.

The metrics used were:

- **Minimal Average Displacement Error (minADE):** The average L2 distance between the predicted and ground-truth paths.
- **Minimal Final Displacement Error (minFDE):** The L2 distance between the predicted and ground-truth final positions in the last time step.

4.1.2 Results and Analysis. The results show that our active learning strategy significantly improves model performance while reducing data requirements. Figure 3 shows the relationship between the percentage of training data used and the performance of the model.

With only 60% of the dataset, our approach achieves 99% of the performance of a model trained on the full dataset, as measured by both minADE and minFDE. Specifically, at 60% data usage, active learning with semantic consistency outperforms random sampling by 20% in minADE and 17% in minFDE. Furthermore, the inclusion of semantic consistency improves performance by 13% compared to active learning without it.

4.2 User Study

The user study aimed to evaluate how natural language explanations affect human interaction with a path prediction system. We focused on three key dimensions: perceived interpretability (assessed with the 7-point Likert question: “**To what extent do you understand the reasons behind the model’s predictions?**”), cognitive load (assessed with the 7-point Likert question: “**How mentally demanding was it to interpret the system decisions?**”), and trust in the model’s outputs (assessed with the 7-point Likert question: “**How much do you trust the model’s predictions?**”). The study employed a within-subject design with three conditions, as depicted in Figure 4: no explanation, presentation of logical rules, and presentation of natural language interpretations of these rules. For the study, we recruited 10 participants (4 female, 6 male) aged 20–31 ($M = 27.2, SD = 3.5$), with varying levels of experience in AI and data annotation. Participants reviewed predicted paths and accompanying explanations (when provided) for 10 scenarios from the Argoverse dataset. Following each condition, participants completed the corresponding questionnaire.

4.2.1 Results. To conduct the statistical analysis of the user study, we performed normality checks using the Shapiro-Wilk test and tested variance homogeneity with Levene’s test. Given the small sample size ($n = 10$) and the ordinal nature of Likert-scale data, we adopted a non-parametric approach to ensure robustness. Specifically, we used the Kruskal-Wallis H test to compare medians across conditions. Post-hoc pairwise comparisons were conducted using Dunn’s test with Bonferroni correction.

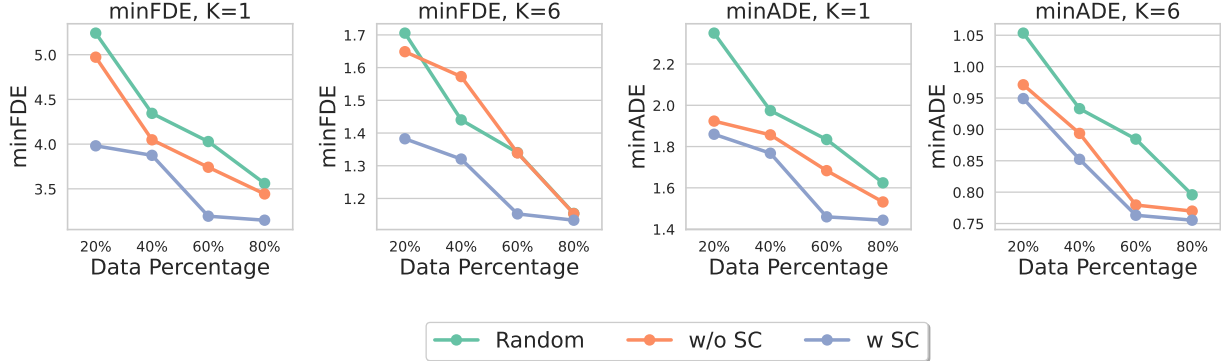


Figure 3: Evaluation of AEXL on the ArgoVerse dataset.

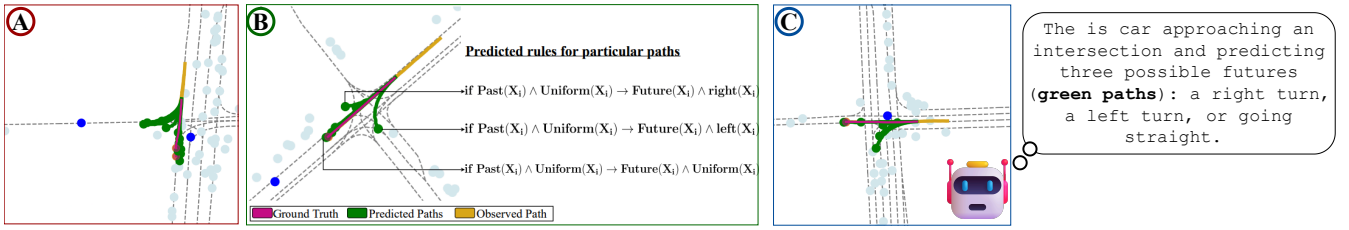


Figure 4: Experimental conditions of the user study: A) represents the “no explanations” condition, B) represents the neuro-symbolic rules as explanations, and C) represents the natural language explanations generated by the LLM.

The results indicate that natural language explanations significantly reduced cognitive load ($Mdn = 2.0$) compared to logical rules ($Mdn = 3.0$, $p < 0.01$) and no explanation ($Mdn = 5.0$, $p < 0.01$). Participants reported the highest interpretability ratings for natural language explanations ($Mdn = 6.5$) compared to logical rules ($Mdn = 4.5$, $p < 0.01$) and no explanation ($Mdn = 3.0$, $p < 0.01$). Trust in the model was also significantly higher with natural language explanations ($Mdn = 6.0$) compared to logical rules ($Mdn = 5.0$, $p < 0.05$) and no explanation ($Mdn = 3.0$, $p < 0.01$).

The preliminary results show that natural language explanations are perceived as more interpretable and trustworthy, and they also reduce cognitive load compared to logical rule-based explanations or no explanations at all.

5 Discussion and Future Work

The results of our preliminary evaluation highlight the effectiveness of AEXL in combining active learning with natural language explanations to enhance interpretability, trust, and data efficiency in path prediction tasks. Our findings demonstrate that integrating neuro-symbolic reasoning with large language models enables the generation of interpretable, human-readable explanations, which significantly improve human-AI interaction during the annotation process. These improvements not only improve annotation quality, but also reduce the cognitive effort required by annotators, fostering a more transparent and collaborative model development process.

Despite these promising results, our work presents some limitations that also pave the way for future exploration. For instance,

the small size of the participant pool in the user study limits the generalizability of the findings to a broader population.

In its current form, the AEXL framework emphasizes uncertainty-based sampling and semantic consistency for active learning. Incorporating additional strategies, such as diversity-based sampling or hybrid uncertainty-diversity approaches, could further improve data efficiency and improve model performance [26]. Investigating alternative scoring metrics for scenario prioritization and annotation will also refine the adaptability of the framework across various path prediction contexts.

Finally, while AEXL successfully combines neuro-symbolic reasoning with LLMs, it does not yet incorporate mechanisms for user feedback to dynamically refine the explanations. Future iterations of the framework could include interactive explanation refinement, allowing annotators to query, clarify, or suggest modifications to the explanations in real time, which would ensure that the explanations align more closely with the annotator needs.

6 Conclusion

In this paper, we introduce AEXL, a novel active learning framework combining neuro-symbolic reasoning and large language models to enhance multimodal path prediction in urban environments. By integrating human-centered explanations and uncertainty-driven sample selection, the framework achieves significant data efficiency, maintaining 99% of a fully supervised model’s performance with only 60% of the training data. The results of our preliminary user study show that the natural language explanations generated by our approach improve user interpretability, trust, and reduce cognitive load. Future work will explore more interaction modalities with the

LLM to provide higher-quality path and explanation corrections, further enhancing the usability and effectiveness of the framework.

References

- [1] Stefano V Albrecht, Cillian Brewitt, John Wilhelm, Balint Gyevnar, Francisco Eiras, Mihai Dobre, and Subramanian Ramamoorthy. 2021. Interpretable goal-based prediction and planning for autonomous driving. In *2021 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 1043–1049.
- [2] Prarthana Bhattacharyya, Chengjie Huang, and Krzysztof Czarnecki. 2022. Ssl-lanes: Self-supervised learning for motion forecasting in autonomous driving. *arXiv preprint arXiv:2206.14116* (2022).
- [3] Samuel Budd, Emma C Robinson, and Bernhard Kainz. 2021. A survey on active learning and human-in-the-loop deep learning for medical image analysis. *Medical Image Analysis* 71 (2021), 102062.
- [4] Holger Caesar, Varun Bankiti, Alex H Lang, Sourabh Vora, Venice Erin Liong, Qiang Xu, Anush Krishnan, Yu Pan, Giancarlo Baldan, and Oscar Beijbom. 2020. nuscenes: A multimodal dataset for autonomous driving. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 11621–11631.
- [5] Erik Cambria, Xulang Zhang, Rui Mao, Melvin Chen, and Kenneth Kwok. 2024. Sentinet 8: Fusing emotion AI and commonsense AI for interpretable, trustworthy, and explainable affective computing. In *International Conference on Human-Computer Interaction*. Springer, 197–216.
- [6] Ming-Fang Chang, John W Lambert, Patsorn Sangkloy, Jagjeet Singh, Slawomir Bak, Andrew Hartnett, De Wang, Peter Carr, Simon Lucey, Deva Ramanan, and James Hays. 2019. Argoverse: 3D Tracking and Forecasting with Rich Maps. In *Conference on Computer Vision and Pattern Recognition (CVPR)*.
- [7] Rohan Chitnis, Tom Silver, Joshua B Tenenbaum, Tomas Lozano-Perez, and Leslie Pack Kaelbling. 2022. Learning neuro-symbolic relational transition models for bilevel planning. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 4166–4173.
- [8] Ivan Donadello, Luciano Serafini, and Artur d'Ávila Garcez. 2017. Logic Tensor Networks for Semantic Image Interpretation. In *Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence, IJCAI-17*. 1596–1602. doi:10.24963/ijcai.2017/221
- [9] Liat Ein Dor, Alon Halfon, Ariel Gera, Eyal Shnarch, Lena Dankin, Leshem Choshen, Marina Danilevsky, Ranit Aharonov, Yoav Katz, and Noam Slonim. 2020. Active learning for BERT: an empirical study. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 7949–7962.
- [10] Achref Doula, Huijie Yin, Max Mühlhäuser, and Alejandro Sanchez Guinea. 2024. NeSyMoF: A Neuro-Symbolic Model for Motion Forecasting. In *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 919–926.
- [11] Yarin Gal, Riashat Islam, and Zoubin Ghahramani. 2017. Deep bayesian active learning with image data. In *International conference on machine learning*. PMLR, 1183–1192.
- [12] Jacob Devlin Ming-Wei Chang Kenton and Lee Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of naacl-HLT*, Vol. 1. Minneapolis, Minnesota, 2.
- [13] Ming Liang, Bin Yang, Rui Hu, Yun Chen, Renjie Liao, Song Feng, and Raquel Urtasun. 2020. Learning lane graph representations for motion forecasting. In *Computer Vision—ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part II* 16. Springer, 541–556.
- [14] Jimpeng Lin, Zhihao Liang, Shengheng Deng, Lile Cai, Tao Jiang, Tianrui Li, Kui Jia, and Xun Xu. 2024. Exploring diversity-based active learning for 3d object detection in autonomous driving. *IEEE Transactions on Intelligent Transportation Systems* (2024).
- [15] Luc de Raedt, Sebastijan Dumančić, Robin Manhaeve, and Giuseppe Marra. 2020. From Statistical Relational to Neuro-Symbolic Artificial Intelligence. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence, IJCAI-20*, Christian Bessière (Ed.). International Joint Conferences on Artificial Intelligence Organization, 4943–4950.
- [16] Burr Settles. 2009. Active learning literature survey. (2009).
- [17] Burr Settles. 2009. Active learning literature survey. (2009).
- [18] Yawar Siddiqui, Julien Valentin, and Matthias Nießner. 2020. Viewal: Active learning with viewpoint entropy for semantic segmentation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 9433–9443.
- [19] Jiankai Sun, Hao Sun, Tian Han, and Bolei Zhou. 2021. Neuro-Symbolic Program Search for Autonomous Driving Decision Module Design. In *Proceedings of the 2020 Conference on Robot Learning (Proceedings of Machine Learning Research, Vol. 155)*, Jens Kober, Fabio Ramos, and Claire Tomlin (Eds.). PMLR, 21–30.
- [20] Stefano Teso and Kristian Kersting. 2019. Explanatory interactive machine learning. In *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society*. 239–245.
- [21] I Tiddi et al. 2020. Neuro-symbolic architectures for context understanding [J]. *Knowledge Graphs for eXplainable Artificial Intelligence: Foundations, Applications and Challenges* 47 (2020), 143.
- [22] Laura Von Rueden, Sebastian Mayer, Katharina Beckh, Bogdan Georgiev, Sven Giesselbach, Raoul Heese, Birgit Kirsch, Julius Pfrommer, Annika Pick, Rajkumar Ramamurthy, et al. 2021. Informed Machine Learning—A taxonomy and survey of integrating prior knowledge into learning systems. *IEEE Transactions on Knowledge and Data Engineering* 35, 1 (2021), 614–633.
- [23] Benjamin Wilson, William Qi, Tanmay Agarwal, John Lambert, Jagjeet Singh, Siddhesh Khandelwal, Bowen Pan, Ratnesh Kumar, Andrew Hartnett, Jhony Kaesemodel Pontes, et al. 2023. Argoverse 2: Next generation datasets for self-driving perception and forecasting. *37th Conference on Neural Information Processing Systems (NeurIPS 2023) Track on Datasets and Benchmarks* (2023).
- [24] Xiaoqian Wu, Yong-Lu Li, Jianhua Sun, and Cewu Lu. 2024. Symbol-LLM: leverage language models for symbolic system in visual human activity reasoning. *Advances in Neural Information Processing Systems* 36 (2024).
- [25] Yi Yang, Zhigang Ma, Feiping Nie, Xiaojun Chang, and Alexander G Hauptmann. 2015. Multi-class active learning by uncertainty sampling with diversity maximization. *International Journal of Computer Vision* 113 (2015), 113–127.
- [26] Xueying Zhan, Huan Liu, Qing Li, and Antoni B Chan. 2021. A comparative survey: Benchmarking for pool-based active learning. In *IJCAI*. 4679–4686.
- [27] Chunhui Zhao, Boao Qin, Shou Feng, Wenxiang Zhu, Weiwei Sun, Wei Li, and Xiuping Jia. 2023. Hyperspectral image classification with multi-attention transformer and adaptive superpixel segmentation-based active learning. *IEEE Transactions on Image Processing* (2023).
- [28] Haiyan Zhao, Hanjie Chen, Fan Yang, Ninghao Liu, Huiqi Deng, Hengyi Cai, Shuaiqiang Wang, Dawei Yin, and Mengnan Du. 2024. Explainability for large language models: A survey. *ACM Transactions on Intelligent Systems and Technology* 15, 2 (2024), 1–38.